Data Structures and Algorithms FINAL

Show equations, substitutions, intermediate reasoning, and a final labeled answer.

1. [6 pts] Trace inserting 7 into the binary-search tree containing 4, 2, 6, 9. List the comparison path and identify the parent of 7.

2. [8 pts] Design an algorithm that detects a duplicate in an array in O(n) expected time. State the data structure, invariant, and space bound.

3. [6 pts] Give tight asymptotic bounds for building a balanced binary-search tree from n=128 sorted keys and for searching it. Explain the construction assumption.

4. [7 pts] Run BFS from A on edges A-B, A-C, B-D, C-E. Give discovery order and explain why BFS finds fewest edges in an unweighted graph.

5. [7 pts] State an insertion-sort loop invariant and explain how initialization, maintenance, and termination establish correctness.

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8. [6 pts] Give tight asymptotic bounds for building a balanced binary-search tree from n=128 sorted keys and for searching it. Explain the construction assumption.

9. [7 pts] Run BFS from A on edges A-B, A-C, B-D, C-E. Give discovery order and explain why BFS finds fewest edges in an unweighted graph.

10. [7 pts] State an insertion-sort loop invariant and explain how initialization, maintenance, and termination establish correctness.